

SIGNVISION

Project Report submitted for the partial fulfilment of grade for the course

MACHINE LEARNING (25SC2107E)

in

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ABSTRACT

The **Real-Time Sign Language Interpreter (SignVision)** is a machine learning-based system designed to recognize hand gestures representing sign language alphabets and convert them into readable text. The system uses a labeled sign language image dataset for training and applies image preprocessing and augmentation to improve model performance. Multiple machine learning approaches, including Logistic Regression, Random Forest, and a Convolutional Neural Network (CNN), can be compared to identify the most effective model for image classification. The trained model is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix. The final model can be integrated into a simple application that accepts images or webcam input and predicts the corresponding sign in real time. The project demonstrates the complete machine learning lifecycle, from data preprocessing and model training to evaluation and deployment, while exploring how machine learning can support communication accessibility for people who use sign language.

Signature of the Guide